

Sumit Mohan

GenAI Engineer | Data Scientist | AI Systems Builder

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Key Highlights:

- Built Astromology.com – GenAI-powered platform using RAG for personalized Astrology
- Founder of Code2Crack – AI-powered EdTech platform with scalable architecture and analytics
- 7+ years of experience in AI/ML, Data Science, and system design
- Expertise in LLMs, RAG pipelines, and production-level AI applications
- Published SCI-indexed research in Machine Learning (Impact Factor: 7)

Professional Summary:

GenAI & Data Science Engineer with 7+ years of experience building scalable AI systems, machine learning models, and intelligent applications.

Expertise in LLMs, Retrieval-Augmented Generation (RAG), and end-to-end AI system design, with hands-on experience in FastAPI, vector databases, and cloud deployment.

Developed Astromology.com, a GenAI-powered platform for kundli analysis and astrology insights, and Code2Crack, an AI-driven EdTech system for adaptive learning and analytics. Strong focus on building production-ready AI solutions from research to deployment.

Technical Skills:

- **GenAI & LLMs:** RAG, Prompt Engineering, LangChain, Transformers, LoRA, Embeddings, Vector Search, Semantic Search
- **Machine Learning:** Scikit-learn, XGBoost, Model Tuning, Feature Engineering
- **Deep Learning:** PyTorch, TensorFlow, NLP
- **Data Science:** Pandas, NumPy, EDA, Data Visualization, Hypothesis Testing
- **Backend:** FastAPI, REST APIs, Microservices
- **Databases:** SQL, NoSQL, Vector Databases (FAISS, Pinecone)
- **Deployment:** Docker, API Deployment, Cloud Basics
- **Programming:** Python (Primary), C++, Java

Projects:

Astromology.com – GenAI-Powered Astrology Platform

- Built and deployed a production-grade GenAI system for personalized astrology insights and analysis
- Designed and implemented a Retrieval-Augmented Generation (RAG) pipeline
- Integrated vector database for semantic search and contextual response generation
- Developed scalable backend using FastAPI and REST APIs
- Applied prompt engineering and embedding strategies to improve response accuracy
- Designed for real-world deployment with scalable architecture and API-based system

Code2Crack – AI-Powered EdTech Platform

- Developed a multi-tenant LMS SaaS platform with AI-driven learning workflows
- Integrated LLM-based tutoring system using retrieval mechanisms
- Built analytics pipelines for student performance tracking
- Designed scalable backend architecture and APIs

Professional Experience:

Founder / Platform Architect ([Code2Crack](#) & [Astromology](#))

- Built a scalable multi-tenant EdTech SaaS platform with AI-driven learning and analytics
- Designed and deployed LLM-based tutoring system using retrieval techniques
- Developed backend APIs and system architecture for scalable deployment
- Built and deployed a production-grade GenAI application ([Astromology](#)) using RAG architecture and LLMs

Head - Technical Training ([CodeQuotient Pvt. Ltd.](#)) | Jan 2025 – Present

- Designed and delivered **data science and machine learning pipelines** including EDA, feature engineering, and model development
- Mentored students on **real-world AI/ML and GenAI projects**, including LLM and RAG-based systems

Senior Technical Trainer ([Sharda University, Greater Noida](#)) | Apr 2023 – Jan 2025

- Delivered training in **Python, Data Science, and Machine Learning** with hands-on project implementation
- Built **end-to-end ML workflows** covering data preprocessing, modeling, and evaluation

Head of Department – CSE ([SDGI Global University, Ghaziabad](#)) | March 2022 – April 2023

- Led curriculum modernization by integrating **AI/ML and data science concepts**
- Strengthened **industry collaboration** for internships and technical training
- Managed academic operations and faculty coordination

Training & Placement ([BIET Jhansi – Autonomous Institute](#)) | March 2019 – February 2022

- Delivered technical training aligned with industry hiring standards
- Mentored students and coordinated internships and placement preparation

Achievements:

- Qualified **UGC-NET and GATE (99 percentile)**
- Published **SCI-indexed research** in Machine Learning
- Built and deployed **AI-based platforms ([Code2Crack](#) & [Astromology](#))**

Publications:

- COVID-19 Impact Prediction using Hybrid ML Models (SCI Indexed)
- Self-Driving Car – Lane Detection (Scopus Indexed)

Educational Qualifications:

- **Ph.D.** in Machine Learning – AKTU, Lucknow (Expected 2026)
- **M.Tech** (2018) | **B.Tech** (2016) | **Polytechnic Diploma** (2013) – Computer Science